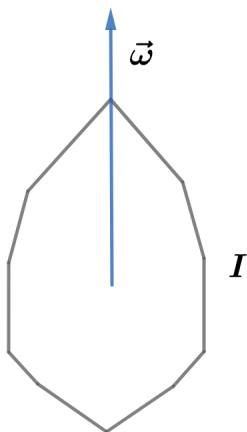


2020B F=ma Exam: Problem 19

Kevin S. Huang



Since there are no external torques, angular momentum is conserved so we have

$$L = I_0\omega_0 = I_f\omega_f$$

If $I_f > I_0$, then $\omega_f < \omega_0$. The kinetic energy is given by

$$T = \frac{L^2}{2I}$$

Since $I_f > I_0$, then $T_f < T_0$. Thus, T and ω have decreased so the answer is A.